

GeeksforGeeks Problem Solving Interface for "Multiply 2 Matrices".

Problem Solved Successfully ✓

Test Cases Passed: **1115 / 1115**

Attempts: Correct / Total: **1 / 1**

Accuracy: 100%

Points Scored: **4 / 4**

Your Total Score: 4 ↑

Time Taken: **0.47**

Compilation Results Custom Input: Y.O.G.I. (AI Bot)

Code (Java):

```
1 import java.util.ArrayList;
2
3 class Solution {
4     public ArrayList<ArrayList<Integer>> multiply(int[][] A, int[][] B) {
5         int n = A.length;
6
7         // Initialize the 2D ArrayList to store the result
8         ArrayList<ArrayList<Integer>> result = new ArrayList<>();
9
10        // Multiply the matrices
11        for (int i = 0; i < n; i++) {
12            // Create a new row for the current iteration
13            ArrayList<Integer> row = new ArrayList<>();
14
15            for (int j = 0; j < n; j++) {
16                int sum = 0;
17                for (int k = 0; k < n; k++) {
18                    sum += A[i][k] * B[k][j];
19                }
20                // Add the computed sum to the current row
21                row.add(sum);
22            }
23
24            // Add the completed row to the final result matrix
25            result.add(row);
26        }
27
28        return result;
29    }
30 }
```

Custom Input Compile & Run Submit

System Tray: 33°C Partly sunny, Search, 11:56 AM 8/11/2026

Problem

Submissions

Leaderboard

Discussions

Editorial

HackerRank

Prepare > Algorithms > Implementation > Matrix Layer Rotation

You are given a 2D matrix of dimension $m \times n$ and a positive integer r . You have to rotate the matrix r times and print the resultant matrix. Rotation should be in anti-clockwise direction.

Rotation of a 4×5 matrix is represented by the following figure. Note that in one rotation, you have to shift elements by one step only.

a_{11}

a_{21}

a_{31}

a_{41}

a_{12}

a_{22}

a_{32}

a_{42}

a_{13}

a_{23}

a_{33}

a_{43}

a_{14}

a_{24}

a_{34}

a_{44}

a_{15}

a_{25}

a_{35}

a_{45}

$\leftarrow$

$\leftarrow$

$\leftarrow$

$\leftarrow$

$\downarrow$

$\downarrow$

$\downarrow$

$\downarrow$

$\leftarrow$

$\leftarrow$

$\rightarrow$

$\rightarrow$

$\uparrow$

$\uparrow$

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$\uparrow$

Matrix Rotation

It is guaranteed that the minimum of m and n will be even.

As an example rotate the Start matrix by 2:

Start

First

Second

57

58

59

60

61

62

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matrix[m - 1 - l][l] = layer[idx]

idx += 1

Left column

for i in range(m - 2 - l, l, -1):

matrix[i][l] = layer[idx]

idx += 1

4. Print the final matrix

for row in matrix:

print(" ".join(map(str, row)))

if __name__ == '__main__':

first_multiple_input = input().rstrip().split()

m = int(first_multiple_input[0])

Line: 82 Col: 1

Run Code

Submit Code

You have earned 80.00 points!

You are now 85 points away from the 3rd star for your problem solving badge.

15%

115/200

Congratulations

You solved this challenge. Would you like to challenge your friends?

Next Challenge

Test case 0

Test case 1

Compiler Message

Success

33°C

Partly sunny

Search

ENG

IN

11:52 AM

8/11/2026

Problem List

1314. Matrix Block Sum

Medium

Topics

Companies

Hint

Given a $m \times n$ matrix `mat` and an integer `k`, return a matrix `answer` where each `answer[i][j]` is the sum of all elements `mat[r][c]` for:

- $i - k \leq r \leq i + k$,
- $j - k \leq c \leq j + k$, and
- (r, c) is a valid position in the matrix.

Example 1:

Input: `mat = [[1,2,3],[4,5,6],[7,8,9]]`
k = 1
Output: `[[12,21,16],[27,45,33],[24,39,28]]`

Example 2:

Input: `mat = [[1,2,3],[4,5,6],[7,8,9]]`
k = 2
Output: `[[45,45,45],[45,45,45],[45,45,45]]`

2.5K 50 1 1 1

Code

Python3 Auto

```
1 class Solution:
2     def matrixBlockSum(self, mat: list[list[int]], k: int) -> list[list[int]]:
3         m, n = len(mat), len(mat[0])
4
5         prefix = [[0] * (n + 1) for _ in range(m + 1)]
6
7         for i in range(1, m + 1):
8             for j in range(1, n + 1):
9                 prefix[i][j] = (mat[i-1][j-1]
```

Saved

Ln 5, Col 9

Test Result

Testcase

Accepted

Runtime: 0 ms

Case 1 Case 2

Input

mat =

[[1,2,3],[4,5,6],[7,8,9]]

k =

1

Output

33°C Partly sunny

Search

ENG IN 11:50 AM 8/11/2026

Problem List

867. Transpose Matrix

Easy

Topics

Companies

Hint

Given a 2D integer array `matrix`, return the *transpose* of `matrix`.

The **transpose** of a matrix is the matrix flipped over its main diagonal, switching the matrix's row and column indices.

1	2	3
4	5	6
7	8	9

 →

1	4	7
2	5	8
3	6	9

Example 1:
Input: `matrix = [[1,2,3],[4,5,6],[7,8,9]]`
Output: `[[1,4,7],[2,5,8],[3,6,9]]`

Example 2:
Input: `matrix = [[1,2,3],[4,5,6]]`

4.2K

133

Star

Refresh

Code

Python

Auto

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14

transposed = [[0] * rows for _ in range(cols)]

for r in range(rows):

for c in range(cols):

transposed[c][r] = matrix[r][c]

return transposed

Saved

Ln 6, Col 9

Test Result

Testcase

Accepted

Runtime: 0 ms

Case 1

Case 2

Input

matrix =

[[1,2,3],[4,5,6],[7,8,9]]

Output

[[1,4,7],[2,5,8],[3,6,9]]

Expected

33°C

Partly sunny

Search

ENG

IN

11:45 AM

8/11/2026

Problem

Submissions

Understand

Discussions

Editorial

← ↻ 🔒 https://www.hackerrank.com/challenges/diagonal-difference/problem?isFullScreen=true

HackerRank

Prepare > Algorithms > Warmup > Diagonal Difference

Exit Full Screen View

Given a square matrix, calculate the absolute difference between the sums of its diagonals.

For example, the square matrix `arr` is shown below:

```
1 2 3
4 5 6
9 8 9
```

- The left-to-right diagonal = $1 + 5 + 9 = 15$.
- The right-to-left diagonal = $3 + 5 + 9 = 17$.

Their absolute difference is $|15 - 17| = 2$.

Function description

Complete the `diagonalDifference` function with the following parameter:

- `int arr[n][m]`: a 2-D array of integers

Return

- `int`: the absolute difference in sums along the diagonals

Input Format

The first line contains a single integer, n , the number of rows and columns in the square matrix `arr`.

Each of the next n lines describes a row, `arr[i]`, and consists of n space-separated integers `arr[i][j]`.

Constraints

- $-100 \leq arr[i][j] \leq 100$

Sample Input

STDIN	Function
3	<code>arr[][]</code> sizes $n = 3$, $m = 3$
11 2 4	<code>arr = [[11, 2, 4], [4, 5, 6], [10, 8, -12]]</code>
4 5 6	
10 8 -12	

Sample Output

15

```
23
24     return abs(primary_sum - secondary_sum)
25
26 if __name__ == '__main__':
27     fptr = open(os.environ['OUTPUT_PATH'], 'w')
28
29     n = int(input().strip())
30
31     arr = []
32
33     for _ in range(n):
34         arr.append(list(map(int, input().rstrip().split())))
35
36     result = diagonalDifference(arr)
37
38     fptr.write(str(result) + '\n')
39
40     fptr.close()
```

Line: 40 Col: 17

Upload Code as File

☐ Test against custom input

Run Code

Submit Code

Problem Solving

You have earned 10.00 points!

You are now 65 points away from the 2nd star for your problem solving badge.

7%

35/100

Congratulations

You solved this challenge. Would you like to challenge your friends?

[f](#) [t](#) [in](#)

Next Challenge

Test case 0

Compiler Message

Success

33°C
Partly sunny

ENG
IN

11:42 AM
8/11/2025

https://leetcode.com/problems/move-zeroes/description/

Problem List

Submit

00:51:35

Premium

Description

Accepted

283. Move Zeroes

Easy

Topics

Companies

Hint

Given an integer array `nums`, move all `0`'s to the end of it while maintaining the relative order of the non-zero elements.

Note that you must do this in-place without making a copy of the array.

Example 1:

Input: `nums = [0,1,0,3,12]`

Output: `[1,3,12,0,0]`

Example 2:

Input: `nums = [0]`

Output: `[0]`

Constraints:

- `1 <= nums.length <= 10^4`

Code

Python

Auto

class Solution(object):
 def moveZeroes(self, nums):
 insert_pos = 0

 for i in range(len(nums)):
 if nums[i] != 0:
 nums[insert_pos], nums[i] = nums[i], nums[insert_pos]
 insert_pos += 1

Saved

Ln 7, Col 13

Test Result

Testcase

Accepted

Runtime: 0 ms

Case 1

Case 2

Input

nums =
[0, 1, 0, 3, 12]

Output

[1, 3, 12, 0, 0]

Expected

33°C
Partly sunny

Search

ENG
IN

11:40 AM
8/11/2026

←↻🔒https://www.hackerrank.com/challenges/time-conversion/problem?isFullScreen=true🔍📄🌟🔖👤⋮🗨️Chat

HackerRank

Prepare > Algorithms > Warmup > Time Conversion

Exit Full Screen View

Problem

Submissions

Leaderboard

Discussions

Editorial

Given a time in 12-hour AM/PM format, convert it to military (24-hour) time.
Note: - 12:00:00AM on a 12-hour clock is 00:00:00 on a 24-hour clock.
- 12:00:00PM on a 12-hour clock is 12:00:00 on a 24-hour clock.

Example

- s = '12:01:00PM'
Return '12:01:00'.
- s = '12:01:00AM'
Return '00:01:00'.

Function Description

Complete the `timeConversion` function with the following parameter(s):

- string s: a time in 12 hour format

Returns

- string: the time in 24 hour format

Input Format

A single string s that represents a time in 12-hour clock format (i.e.: hh:mm:ssAM or hh:mm:ssPM).

Constraints

- All input times are valid

Sample Input 0

07:05:45PM

Sample Output 0

19:05:45

12:00:00

```
1#!/bin/python3
2import math
3import os
4import random
5import re
6import sys
7
8#
9# Complete the 'timeConversion' function below.
10#
11# The function is expected to return a STRING.
12# The function accepts STRING s as parameter.
13#
14
15def timeConversion(s):
16    # Extract the AM/PM period
17    period = s[-2:]
18
19    # Extract the hour as an integer
20    hour = int(s[:2])
21
22    # Extract the minutes and seconds
23    minutes_seconds = s[2:-2]
24
25    if period == 'AM':
26        # 12 AM translates to 00 in 24-hour format
27        if hour == 12:
28            hour = 0
29
30    else: # PM
31        # For PM, add 12 to the hours, except for 12 PM which stays 12
32        if hour != 12:
33            hour += 12
34
35    # Format the hour to always be two digits (e.g., 0 becomes 00, 7 becomes 07)
36    # and append the rest of the time string.
37    return f"{hour:02}{minutes_seconds}"
```

Line: 48 Col: 1

Upload Code as File

Test against custom input

Run Code

Submit Code

33°C
Partly sunny

🔍 Search

ENG
IN 🔊 🔌 🔋 11:38 AM
8/11/2026

Problem List

14.2K 466

Accepted

Editorial

Description

Explanation

The element 1 occurs at the indices 0 and 3.

Example 2:

Input: nums = [1,2,3,4]

Output: false

Explanation:

All elements are distinct.

Example 3:

Input: nums = [1,1,1,3,3,4,3,2,4,2]

Output: true

Constraints:

$1 \leq \text{nums.length} \leq 10^5$

$-10^9 \leq \text{nums}[i] \leq 10^9$

Code

Python Auto

```
1 class Solution(object):
2     def containsDuplicate(self, nums):
3         return len(set(nums)) != len(nums)
4
```

Saved Ln 3, Col 43

Test Result

Testcase

Accepted Runtime: 0 ms

Case 1

Case 2

Case 3

Input

nums = [1,2,3,1]

Output

true

Expected

33°C Partly sunny

Search

ENG IN 11:33 AM 8/11/2026

Problem

Submissions

Leaderboard

Discussions

33°C
Partly sunny

Search

ENG
IN

11:28 AM
8/11/2026

https://www.hackerrank.com/challenges/compare-the-triplets/problem?isFullScreen=true

Exit Full Screen View

HackerRank

Prepare > Algorithms > Warmup > Compare the Triplets

Alice and Bob each created one problem for HackerRank. A reviewer rates the two challenges, awarding points on a scale from 1 to 100 for three categories: problem clarity, originality, and difficulty.

The rating for Alice's challenge is the triplet $a = (a[0], a[1], a[2])$, and the rating for Bob's challenge is the triplet $b = (b[0], b[1], b[2])$.

The task is to calculate their comparison points by comparing each category:

- If $a[i] > b[i]$, then Alice is awarded 1 point.
- If $a[i] < b[i]$, then Bob is awarded 1 point.
- If $a[i] = b[i]$, then neither person receives a point.

Example

$a = [1, 2, 3]$
 $b = [3, 2, 1]$

- For elements $a[0]$, Bob is awarded a point because $a[0] < b[0]$.
- For the equal elements $a[1]$ and $b[1]$, no points are earned.
- Finally, for elements $a[2]$, $a[2] > b[2]$ so Alice receives a point.

The return array is $[1, 1]$ with Alice's score first and Bob's second.

Function Description

Complete the function `compareTriplets` with the following parameter(s):

- `int a[3]`: Alice's challenge rating
- `int b[3]`: Bob's challenge rating

Returns

- `int[2]`: the first element is Alice's score and the second is Bob's score

30

```
if __name__ == '__main__':  
    fptr = open(os.environ['OUTPUT_PATH'], 'w')  
    a = list(map(int, input().rstrip().split()))  
    b = list(map(int, input().rstrip().split()))  
    result = compareTriplets(a, b)  
    fptr.write(str(result) + '\n')  
    fptr.close()
```

Line: 43 Col: 1

Upload Code as File

Test against custom input

Run Code

Submit Code

Problem Solving

You have earned 10.00 points!

You are now 20 points away from the 1st star for your problem solving badge.

33%

10/30

Congratulations

You solved this challenge. Would you like to challenge your friends?

[f](#) [t](#) [in](#)

Next Challenge

Test case 0

Compiler Message

Test case 1

Success

Lapindromes

Lapindrome is defined as a string which when split in the middle, gives two halves having the same characters and same frequency of each character. If there are odd number of characters in the string, we ignore the middle character and check for lapindrome. For example **gaga** is a lapindrome, since the two halves **ga** and **ga** have the same characters with same frequency. Also, **abccab**, **rotor** and **xyzxy** are a few examples of lapindromes. Note that **abbaab** is NOT a lapindrome. The two halves contain the same characters but their frequencies do not match.

Your task is simple. Given a string, you need to tell if it is a lapindrome.

Input:

First line of input contains a single integer **T**, the number of test cases.

Each test is a single line containing a string **S** composed of only lowercase English alphabet.

Output:

For each test case, output on a separate line: "YES" if the string is a lapindrome and "NO" if it is not.

Constraints:

- $1 \leq T \leq 100$
- $2 \leq |S| \leq 1000$, where $|S|$ denotes the length of **S**

Sample 1:

```
20-         else:
21-             right_half = s[mid+1:]
22-
23-
24-         if sorted(left_half) == sorted(right_half):
25-             print("YES")
26-         else:
27-             print("NO")
28-
29- if __name__ == '__main__':
30-     solve()
```

Test against Custom Input

```
rotor
xyzxy
abbaab
ababc
```

✓ Correct Answer

Submission ID: [1338208911](#)

Sub-Task	Task #	Result (time)
1	0	Correct (0.01)

Visualize Code

Run

Submit

33°C
Partly sunny



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Problem List

1704. Determine if String Halves Are Alike

Easy Topics Companies Hint

You are given a string `s` of even length. Split this into two halves of equal lengths, and let `a` be the first half and `b` be the second half.

Two strings are **alike** if they have the same number of vowels ('a', 'e', 'i', 'o', 'u', 'A', 'E', 'I', 'O', 'U'). Notice that `s` contains uppercase and lowercase letters.

Return `true` if `a` and `b` are **alike**. Otherwise, return `false`.

Example 1:

Input: `s = "book"`

Output: `true`

Explanation: `a = "bo"` and `b = "ok"` has 1 vowel and `b` has 1 vowel. Therefore, they are alike.

Example 2:

Input: `s = "leetcode"`

Output: `false`

Explanation: `a = "leet"` and `b = "code"` has 3 vowels and `b` has 2 vowels. Therefore, they are not alike.

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Code

Python Auto

```
1 class Solution(object):
2     def halvesAreAlike(self, s):
3         vowels = set('aeiouAEIOU')
4         mid = len(s) // 2
5
6
7         count_a = sum(1 for char in s[:mid] if char in vowels)
8
9         count_b = sum(1 for char in s[mid:] if char in vowels)
10
11        return count_a == count_b
```

Saved

Ln 8, Col 5

Test Result

Testcase

Accepted Runtime: 0 ms

Case 1

Case 2

Input

`s =`

"book"

Output

true

Expected

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